

Severity-resolved reanalysis of the aging mouse auditory system

public microarrays and single-nucleus RNA-seq; all 2,019 CellChat mouse interactions

Datasets and design

Cochlea

GSE49543 | 41 samples
Affymetrix MOE430A

Inferior colliculus

GSE49522 | 39 samples
same strain and severity scale

Severity model

young control -> middle age
-> mild -> severe presbycusis

Single-nucleus RNA-seq

GSE274279 | 11,403 nuclei
3 / 12 / 24 months
147 macrophages / microglia

Validation

GSE233798, GSE153882,
GSE154833,
nested cross-validation

Analysis

limma / eBayes severity and pairwise models



WGCNA module M12: 66 severity genes



Peripheral-central convergence: 48 shared genes



Marker-based immune deconvolution



Single-nucleus myeloid localization



TF, pathway and drug-signature enrichment



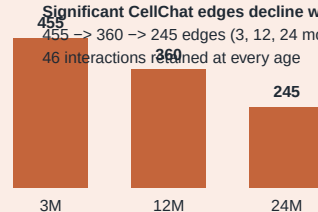
CellChat across all 2,019 mouse interactions

Key findings

Significant CellChat edges decline with age

455 -> 360 -> 245 edges (3, 12, 24 months)

46 interactions remained at every age



PTN signaling is the strongest pathway at all ages

PTPRC-MRC1 myeloid self-communication at every age

APP-CD74 signaling emerges from 12 months

Sex adjustment preserves the myeloid direction

External transfer of the model remains limited

Convergent macrophage / microglial program in presbycusis